**PAST EXAMINATION QUESTIONS (FP2)**

**TOPIC 16: SECOND ORDER DIFFERENTIAL EQUATIONS**

**1.** The variables $x$ and $y$ are related by the differential equation

$$9\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + y = 3x^2 + 30x.$$

(a) Find the general solution for $y$ in terms of $x$. $\left(\boldsymbol{y = e^{-\frac{1}{3}x}(Ax + B) + 3x^2 - 6x - 18}\right)$ **[6]**

(b) State an approximation solution for large positive values of $x$. $(\boldsymbol{y = 3x^2 - 6x - 18})$ **[1]**

**Oct/Nov 2020/21/23/2**

**2.** Find the particular solution of the differential equation

$$\frac{d^2x}{dt^2} + 8\frac{dx}{dx} + 15x = 102\cos 3t,$$

given that, when $t = 0, x = 1$ and $\frac{dx}{dt} = 0$. $(\boldsymbol{x = 6e^{-5t} - 6e^{-3t} + 4\sin 3t + \cos 3t})$ **[11]**

**Oct/Nov 2020/22/6**

**3.** It is given that $x = t^3y$ and

$$t^3\frac{d^2y}{dt^2} + (4t^3 + 6t^2)\frac{dy}{dt} + (13t^3 + 12t^2 + 6t)y = 61e^{\frac{1}{2}t}.$$

(a) Show that

$$\frac{d^2x}{dt^2} + d\frac{dx}{dt} + 13x = 61e^{\frac{1}{2}t}.$$

**[4]**

(b) Find the general solution for $y$ in terms of $t$. $\left(\boldsymbol{y = t^{-3}e^{-2t}(A\cos 3t + B\sin 3t) + 4t^{-3}e^{\frac{1}{2}t}}\right)$ **[7]**

**May/June 2020/21/22/7**

**4.** It is given that $w = \cos y$ and

$$\tan y\,\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 + 2\tan y\frac{dy}{dx} = 1 + e^{-2x}\sec y.$$

(i) Show that

$$\frac{d^2w}{dx^2} + 2\frac{dw}{dx} + w = -e^{-2x}.$$

**[4]**

(ii) Find the particular solution for $y$ in terms of $x$, given that when $x = 0, y = \frac{1}{3}\pi$ and $\frac{dy}{dx} = \frac{1}{\sqrt{3}}$. $(\boldsymbol{y - e^{-2x}})$ **[10]**

**Oct/Nov 2019/11/12/13/11E**

**5.** Find the particular solution of the differential equation

$$10\frac{d^2x}{dt^2}+3\frac{dx}{dt}-x=t+2,$$

given that when $t=0, x=0$ and $\frac{dx}{dt}=0$. $\left(\boldsymbol{x=5e^{\frac{1}{5}t}-t-5}\right)$ **[10]**

**May/June 2019/11/12/7**

**6.** Find the particular solution of the differential equation

$$9\frac{d^2x}{dt^2}+6\frac{dx}{dt}+x=50\sin t,$$

given that when $t=0, x=0$ and $\frac{dx}{dt}=0$. $\left(\boldsymbol{x=(5t+3)e^{-\frac{1}{3}t}-4\sin t-3\cos t}\right)$ **[10]**

**May/June 2019/13/8**

**7.** (i) Find the particular solution of the differential equation

$$\frac{d^2x}{dt^2}+2\frac{dx}{dt}+10x=37\sin 3t,$$

given that $x=3$ and $\frac{dx}{dt}=0$ when $t=0$. $(\boldsymbol{x=e^{-t}(9\cos 3t+2\sin 3t)-6\cos 3t+\sin 3t})$ **[10]**

(ii) Show that, for large positive values of $t$ and for any initial conditions,

$$x\approx\sqrt{37}\sin(3t-\phi),$$

where the constant $\phi$ is such that $\tan\phi=6$. **[3]**

**Oct/Nov 2018/11/13/10**

**8.** (i) Find the general solution of the differential equation

$$\frac{d^2x}{dt^2}+2\frac{dx}{dt}+x=4\sin t.$$ **[7]**

$(\boldsymbol{x=(A+Bt)e^{-t}-2\cos t})$

(ii) State an approximation solution for large positive values of $t$. $(\boldsymbol{x\approx -2\cos t})$ **[1]**

**Oct/Nov 2018/12/4**

**9.** Find the particular solution of the differential equation

$$49\frac{d^2y}{dx^2}+14\frac{dy}{dx}+y=49x+735,$$

given that when $x=0, y=0$ and $\frac{dy}{dx}=0$. $\left(\boldsymbol{y=-(49+56x)e^{-\frac{1}{7}x}+49(x+1)}\right)$ **[10]**

**May/June 2018/11/12/7**

**10.** It is given that $t \neq 0$ and

$$t\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + 9tx = 3t^2 + 1.$$

(i) Show that if $y = tx$ when

$$\frac{d^2y}{dt^2} + 9y = 3t^2 + 1.$$ **[3]**

(ii) Find $x$ in terms of $t$, given that $x = \frac{1}{9}\pi$ and $\frac{dx}{dt} = \frac{2}{3}$ when $t = \frac{1}{3}\pi$. $\left(\boldsymbol{x = \frac{\cos 3t - \pi \sin 3t + 9t^2 + 1}{27t}}\right)$ **[9]**

**May/June 2018/13/10**

**11.** Find the general solution of the differential equation

$$\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + 5x = 4 - 5t^2.$$ **[6]**

$$\left(\boldsymbol{x = e^{-t}(A\cos 2t + B\sin 2t) + \frac{22}{25} + \frac{4}{5}t - t^2}\right)$$

**Oct/Nov 2017/11/12/13/2**

**12.** It is given that $x = t^{\frac{1}{2}}$, where $x > 0$ and $t > 0$, and $y$ is a function of $x$.

(i) Show that

$$\frac{dy}{dx} = 2t^{\frac{1}{2}}\frac{dy}{dt} \text{ and } \frac{d^2y}{dx^2} = 2\frac{dy}{dt} + 4t\frac{d^2y}{dt^2}.$$ **[3]**

(ii) Hence show that the differential equations

$$\frac{d^2y}{dx^2} - \left(8x + \frac{1}{x}\right)\frac{dy}{dx} + 12x^2y = 4x^2e^{-x^2} \qquad (*)$$

reduces to the differential equation

$$\frac{d^2y}{dt^2} - 4\frac{dy}{dt} + 3y = e^{-t}.$$ **[1]**

(iii) Find the general solution of (*), giving $y$ in terms of $x$. $\left(\boldsymbol{y = Ae^{x^2} + Be^{3x^2} + \frac{1}{8}e^{-x^2}}\right)$ **[7]**

**May/June 2017/11/12/10**

**13.** Find the solution of the differential equation

$$\frac{d^2x}{dt^2} + 6\frac{dx}{dt} + 9x = 18t^2 + 6t + 1,$$

given that when $t = 0, x = 3$ and $\frac{dx}{dt} = 0$. $\left(\boldsymbol{x = 2e^{-3t} + 8te^{-3t} + 2t^2 - 2t + 1}\right)$ **[10]**

**May/June 2017/13/8**

**14.** Find the general solution of the differential equation

$$\frac{d^2x}{dt^2} + 7\frac{dx}{dt} + 10x = 116\sin 2t.$$ **[8]**

State an approximate solution for large positive values of $t$. ($\boldsymbol{x = Ae^{-2t} + Be^{-5t} + 3\sin 2t - 7\cos 2t}$; $\boldsymbol{3\sin 2t - 7\cos 2t}$) **[1]**

**Oct/Nov 2016/11/12/13/6**

**15.** Find the value of the constant $k$ such that $y = kx^2e^{2x}$ is a particular integral of the differential equation

$$\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = 4e^{2x}. \quad (*)$$ **[4]**

($\boldsymbol{k = 2}$)

Hence find the general solution of (*). $\left(\boldsymbol{y = Ae^{2x} + Bxe^{2x} + 2x^2e^{2x}}\right)$ **[3]**

Find the particular solution of (*) such that $y = 3$ and $\frac{dy}{dx} = -2$ when $x = 0$. $\left(\boldsymbol{y = 3e^{2x} - 8xe^{2x} + 2x^2e^{2x}}\right)$ **[4]**

**May/June 2016/11/12/9**

**16.** Given that $y$ is a function of $x$ and that $x = e^u$, show that

$$x\frac{dy}{dx} = \frac{dy}{du} \quad and \quad x^2\frac{d^2y}{dx^2} = \frac{d^2y}{du^2} - \frac{dy}{du}.$$ **[3]**

Given also that

$$x^2\frac{d^2y}{dx^2} + 3x\frac{dy}{dx} + 17y = 34\ln x + 21,$$

deduce that

$$\frac{d^2y}{du^2} + 2\frac{dy}{du} + 17y = 34u + 21.$$ **[1]**

Find $y$ in terms of $x$ given that $y = 0$ and $\frac{dy}{dx} = -1$ when $x = 1$. $\left(\boldsymbol{y = 2\ln x + 1 - \frac{1}{x}[\cos(4\ln x) + \sin(4\ln x)]}\right)$ **[9]**

**May/June 2016/13/10**

**17.** Find the general solution of the differential equation

$$\frac{d^2x}{dt^2} + 4\frac{dx}{dt} + 4x = 7 - 2t^2.$$ **[6]**

$$\left(\boldsymbol{x = Ae^{-2t} + Bte^{-2t} - \frac{1}{2}t^2 + t + 1}\right)$$

**Oct/Nov 2015/11/12/13/2**

**18.** Show that the substitution $v = \frac{1}{y}$ reduces the differential equation

$$\frac{2}{y^3}\left(\frac{dy}{dx}\right)^2 - \frac{1}{y^2}\frac{d^2y}{dx^2} - \frac{2}{y^2}\frac{dy}{dx} + \frac{5}{y} = 17 + 6x - 5x^2$$

to the differential equation

$$\frac{d^2v}{dx^2} + 2\frac{dv}{dx} + 5v = 17 + 6x - 5x^2.$$ **[4]**

Hence find $y$ in terms of $x$, given that when $x = 0, y = \frac{1}{2}$ and $\frac{dy}{dx} = -1$. **[10]**

$$\left(\boldsymbol{y = \left[e^{-x}\left(\frac{1}{2}\sin 2x - \cos 2x\right) + 3 + 2x - x^2\right]^{-1}}\right)$$

**May/June 2015/11/12/11E**

**19.** Find the particular solution of the differential equation

$$\frac{d^2x}{dt^2} - 3\frac{dx}{dt} - 10x = 2\sin t - 3\cos t,$$

given that, when $t = 0, x = 3.3$ and $\frac{dx}{dt} = 0.9$. $\left(\boldsymbol{x = e^{5t} + 2e^{-2t} + 0.3\cos t - 0.1\sin t}\right)$ **[11]**

**May/June 2015/13/9**

**20.** Given that

$$x\frac{d^2y}{dx^2} + (2x + 2)\frac{dy}{dx} + (2 - 3x)y = 10e^{2x}$$

and that $v = xy$, show that

$$\frac{d^2v}{dx^2} + 2\frac{dv}{dx} - 3v = 10e^{2x}.$$ **[4]**

Find the general solution for $y$ in terms of $x$. $\left(\boldsymbol{y = \frac{1}{x}\left(Ae^{-3x} + Be^{x} + 2e^{2x}\right)}\right)$ **[7]**

**Oct/Nov 2014/11/12/13/9**

**21.** Obtain the general solution of the differential equation

$$\frac{d^2x}{dt^2} - 6\frac{dx}{dt} + 25x = 195\sin 2t.$$ **[6]**

$$(\boldsymbol{x = e^{3t}(A\cos 4t + B\sin 4t) + 7\sin 2t + 4\cos 2t})$$

**May/June 2014/11/12/4**

**19.** Find the particular solution of the differential equation

$$\frac{d^2x}{dt^2} + 0.16\frac{dx}{dt} + 0.0064x = 8.64 + 0.32t,$$

given that when $t = 0, x = 0$ and $\frac{dx}{dt} = 0$. $\left(\boldsymbol{x = 100 + 50t - (100 + 58t)e^{-0.08t}}\right)$ **[10]**

Show that, for large positive $t$, $\frac{dx}{dt} \approx 50$. **[2]**

**May/June 2014/13/10**

**22.** Given that

$$y^2\frac{d^2y}{dx^2} - 6y^2\frac{dy}{dx} + 2y\left(\frac{dy}{dx}\right)^2 + 3y^3 = 25e^{-2t}$$

and that $v = y^3$, show that

$$\frac{d^2v}{dx^2} - 6\frac{dv}{dx} + 9v = 75e^{-2x}.$$ **[4]**

Find the particular solution for $y$ in terms of $x$, given that when $x = 0, y = 2$ and $\frac{dy}{dx} = 1$. **[10]**

$\left(\boldsymbol{y = \left[5e^{3x} + 3xe^{3x} + 3e^{-2x}\right]^{\frac{1}{3}}}\right)$

**Oct/Nov 2013/11/12/11O**

**23.** Find the general solution of the differential equation

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 4y = 4x^2 + 8.$$ **[7]**

$\left(\boldsymbol{y = e^{-x}\left(A\cos\sqrt{3}\,x + B\sin\sqrt{3}\,x\right) + x^2 - x + 2}\right)$

**Oct/Nov 2013/13/3**

**24.** Find $x$ in terms of $t$ given that

$$4\frac{d^2x}{dt^2} + 4\frac{dx}{dt} + x = 6e^{-2t},$$

and that, when $t = 0, x = \frac{5}{3}$ and $\frac{dx}{dt} = \frac{7}{6}$. $\left(\boldsymbol{x = e^{-\frac{t}{2}} + 3te^{-\frac{t}{2}} + \frac{2}{3}e^{-2t}}\right)$ **[9]**

State $\lim_{t\to\infty} x$. $(\boldsymbol{0})$ **[1]**

**May/June 2013/11/12/9**

**25.** Find the value of the constant $\lambda$ such that $\lambda xe^{-x}$ is a particular integral of the differential equation

$$\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 4y = 6e^{-x}.$$ **[4]**

$(\boldsymbol{\lambda = 2})$

Find the solution of the differential equation for which $y = 2$ and $\frac{dy}{dx} = 3$ when $x = 0$. $(\boldsymbol{y = 3e^{-x} - e^{-4x} + 2xe^{-x}})$ **[6]**

**May/June 2013/13/7**

**26.** Find the general solution of the differential equation

$$\frac{d^2x}{dt^2} + 4\frac{dx}{dt} + 13x = 26t^2 + 3t + 13.$$ **[6]**

$(\boldsymbol{x = e^{-2t}(A\cos 3t + B\sin 3t) + 2t^2 - t + 1})$

**Oct/Nov 2012/11/12/3**

**27.** Obtain the general solution of the differential equation

$$\frac{d^2x}{dt^2} + 6\frac{dx}{dt} + 13x = 75\cos 2t.$$ **[7]**

$\left(\boldsymbol{x = e^{-3t}(A\cos 2t + B\sin 2t) + 3\cos 2t + 4\sin 2t}\right)$

Given that $x = 5$ and $\frac{dx}{dt} = 0$ when $t = 0$, find $x$ in terms of $t$. **[4]**

$\left(\boldsymbol{x = e^{-3t}(2\cos 2t - \sin 2t) + 3\cos 2t + 4\sin 2t}\right)$

Show that, for large positive values of $t$ and for any initial conditions,

$$x \approx 5\cos(2t - \phi),$$

where the constant $\phi$ is such that $\tan\phi = \frac{4}{3}$. **[3]**

**Oct/Nov 2012/13/12O**

**28.** Find the particular solution of the differential equation

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 5y = 10e^{-2x},$$

given that $y = 5$ and $\frac{dy}{dx} = 1$ when $x = 0$. $\left(\boldsymbol{y = e^{-x}(3\cos 2x + 4\sin 2x) + 2e^{-2x}}\right)$ **[11]**

**May/June 2012/11/12/8**

**29.** Show that the substitution $y = xz$ reduces the differential equation

$$\frac{1}{x}\frac{d^2y}{dx^2} + \left(\frac{6}{x} - \frac{2}{x^2}\right)\frac{dy}{dx} + \left(\frac{9}{x} - \frac{6}{x^2} + \frac{2}{x^3}\right)y = 169\sin 2x$$

to the differential equation

$$\frac{d^2z}{dx^2} + 6\frac{dz}{dx} + 9z = 169\sin 2x.$$ **[4]**

Find the particular solution for $y$ in terms of $x$, given that when $x = 0, z = -10$ and $\frac{dz}{dx} = 5$. **[10]**

$(\boldsymbol{y = 2xe^{-3x} + x^2e^{-3x} + 5x\sin 2x - 12x\cos 2x})$

**May/June 2012/13/11O**

**30.** Given that

$$x^2\frac{d^2y}{dx^2} + 4x(1+x)\frac{dy}{dx} + 2(1+4x+2x^2)y = 8x^2$$

and that $x^2y = z$, show that

$$\frac{d^2z}{dx^2} + 4\frac{dz}{dx} + 4z = 8x^2.$$ **[4]**

Find the general solution for $y$ in terms of $x$. $\left(\boldsymbol{y = \frac{A}{x^2}e^{-2x} + \frac{B}{x}e^{-2x} + 2 - \frac{4}{x} + \frac{3}{x^2}}\right)$ **[8]**

Describe the behaviour of $y$ as $x \to \infty$. $(\boldsymbol{y \to 2})$ **[2]**

**Oct/Nov 2011/11/12/11O**

**31.** Find the general solution of the differential equation

$$\frac{d^2x}{dt^2} + 4\frac{dx}{dt} + 4x = \sin 2t.$$ **[6]**

$\left(\boldsymbol{x = Ae^{-2t} + Bte^{-2t} - \frac{1}{8}\cos 2t}\right)$

Describe the behaviour of $x$ as $t \to \infty$, justifying your answer. $\left(\boldsymbol{x \approx -\frac{1}{8}\cos 2t\,; x \text{ oscillates}}\right)$ **[2]**

**Oct/Nov 2011/13/6**

**32.** The variables $x$ and $y$ are related by the differential equation

$$y^2\frac{d^2y}{dx^2} + 2y^2\frac{dy}{dx} + 2y\left(\frac{dy}{dx}\right)^2 - 5y^3 = 8e^{-x}.$$

Given that $v = y^3$, show that

$$\frac{d^2v}{dx^2} + 2\frac{dv}{dx} - 15v = 24e^{-x}.$$ **[4]**

Hence find the general solution for $y$ in terms of $x$. $\left(\boldsymbol{y = \left\{Ae^{-5x} + Be^{3x} - \frac{3}{2}e^{-x}\right\}^{\frac{1}{3}}}\right)$ **[7]**

**May/June 2011/11/12/7**

**33.** Find the general solution of the differential equation

$$\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + 5x = 10\sin t.$$ **[6]**

$(\boldsymbol{x = e^{-t}(A\cos 2t + B\sin 2t) + 2\sin t - \cos t})$

Find the particular solution, given that $x = 5$ and $\frac{dx}{dt} = 2$ when $t = 0$. **[4]**

$(\boldsymbol{x = e^{-t}(6\cos 2t + 3\sin 2t) + 2\sin t - \cos t})$

State an approximate solution for large positive values of $t$. $(\boldsymbol{x \approx 2\sin t - \cos t})$ **[1]**

**May/June 2011/13/8**

**34.** It is given that $x \neq 0$ and

$$x\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 4xy = 8x^2 + 16.$$

Show that if $z = xy$ then

$$\frac{d^2z}{dx^2} + 4z = 8x^2 + 16.$$ **[3]**

Find $y$ in terms of $x$, given that $y = 0$ and $\frac{dy}{dx} = -2$ when $x = \frac{1}{2}\pi$. **[6]**

$$\left(\boldsymbol{y = \frac{1}{x}\left[\left(\frac{\pi^2}{2} + 3\right)\cos 2x + \frac{3\pi}{2}\sin 2x + 2x^2 + 3\right]}\right)$$

**Oct/Nov 2010/1/11**

**35.** The variables $z$ and $x$ are related by the differential equation

$$3z^2\frac{d^2z}{dx^2} + 6z^2\frac{dz}{dx} + 6z\left(\frac{dz}{dx}\right)^2 + 5z^3 = 5x + 2.$$

Use the substitution $y = z^3$ to shoe that $y$ and $x$ are related by the differential equation

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 5y = 5x + 2.$$ **[3]**

Given that $z = 1$ and $\frac{dz}{dx} = -\frac{2}{3}$ when $x = 0$, find $z$ in terms of $x$. $\left(\boldsymbol{z = [e^{-x}(\cos 2x - \sin 2x) + x]^{\frac{1}{3}}}\right)$ **[9]**

Deduce that, for larger positive values of $x$, $z \approx x^{\frac{1}{3}}$. **[2]**

**May/June 2010/11/12/11E**

**36.** Obtain the general solution of the differential equation

$$\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 4y = 10\sin 3x - 20\cos 3x.$$ **[5]**

$(\boldsymbol{y = Ae^{-x} + Be^{-4x} - 1.4\sin 3x - 0.2\cos 3x})$

Show that, for large positive $x$ and independently of the initial conditions,

$$y \approx R\sin(3x + \phi),$$

where the constants $R$ and $\phi$, such that $R > 0$ and $0 < \phi < 2\pi$, are to be determined correct to 2 decimal places. $(\boldsymbol{R = \sqrt{2}, \phi = 3.28})$ **[4]**

**May/June 2010/13/8**

**37.** Show that if $y$ depends on $x$ and $x = e^u$ then

$$x^2\frac{d^2y}{dx^2} = \frac{d^2y}{du^2} - \frac{dy}{du}.$$ **[4]**

Given that $y$ satisfies the differential equation

$$x^2\frac{d^2y}{dx^2} + 5x\frac{dy}{dx} + 3y = 30x^2,$$

use the substitution $x = e^u$ to show that

$$\frac{d^2y}{du^2} + 4\frac{dy}{du} + 3y = 30e^{2u}.$$ **[2]**

Hence find the general solution for $y$ in terms of $x$. $\left(\boldsymbol{y = \frac{A}{x} + \frac{B}{x^3} + 2x^2}\right)$ **[5]**

**Oct/Nov 2009/1/9**

**38.** Find the general solution of the differential equation

$$4\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 65y = 65x^2 + 8x + 73.$$ **[6]**

Show that, whatever the initial conditions, $\frac{y}{x^2} \to 1$ as $x \to \infty$. $\left(\boldsymbol{y = e^{-\frac{x}{2}}(A\sin 4x + B\cos 4x) + x^2 + 1}\right)$ **[2]**

**May/June 2009/1/8**

**39.** Find $y$ in terms of $t$, given that

$$5\frac{d^2y}{dt^2}+6\frac{dy}{dt}+5y=15+12t+5t^2,$$

and that $y=\frac{dy}{dt}=0$ when $t=0$. $\left(\boldsymbol{y=-\frac{1}{4}e^{-\frac{3t}{5}}\left[4\cos\left(\frac{4t}{5}\right)+3\sin\left(\frac{4t}{5}\right)\right]+t^2+1}\right)$ **[9]**

**Oct/Nov 2008/1/8**

**40.** Show that, with a suitable value of the constant $\alpha$, the substitution $y=x^{\alpha}w$ reduces the differential equation

$$2x^2\frac{d^2y}{dx^2}+(3x^2+8x)\frac{dy}{dx}+(x^2+6x+4)y=f(x)$$

to

$$2\frac{d^2w}{dx^2}+3\frac{dw}{dx}+w=f(x).$$ **[5]**

Find the general solution for $y$ in the case where $f(x)=6\sin 2x+7\cos 2x$. **[6]**

$\left(\boldsymbol{\alpha=2;\, y=x^{-2}\left(Ae^{-x}+Be^{-\frac{x}{2}}-\cos 2x\right)}\right)$

**May/June 2008/11/1**

**41.** Show that the substitution $y=\frac{1}{w}$ reduces the differential equation

$$y\frac{d^2y}{dx^2}+2y\frac{dy}{dx}-2\left(\frac{dy}{dx}\right)^2-5y^2=(5x^2+4x+2)y^3$$

to

$$\frac{d^2w}{dx^2}+2\frac{dw}{dx}+5w=-5x^2-4x-2.$$ **[4]**

Find the general solution for $w$ in terms of $x$. $\left(\boldsymbol{w=e^{-x}(A\cos 2x+B\sin 2x)-x^2}\right)$ **[6]**

**[3]**

Find a function f such that $\lim_{x\to\infty}\left(\frac{y}{f(x)}\right)=1$. $\left(\boldsymbol{f(x)=-\frac{1}{x^2}}\right)$

**Oct/Nov 2007/1/12O**

**42.** Find the general solution of the differential equation

$$\frac{d^2y}{dx^2}+4\frac{dy}{dx}+29y=58x+37.$$ **[6]**

$\left(\boldsymbol{y=e^{-2x}(A\cos 5x+B\sin 5x)+2x+1}\right)$

**May/June 2007/1/3**

**43.** Given that

$$2x^3\frac{d^2y}{dx^2}+12y^3\frac{dy}{dx}+6y^2\left(\frac{dy}{dx}\right)^2+17y^4=13e^{-4x}$$

and that $v=y^4$, show that

$$\frac{d^2v}{dx^2}+6\frac{dv}{dx}+34v=26e^{-4x}.$$ **[4]**

Hence find the general solution for $y$ in terms of $x$. $\left(\boldsymbol{y=\{e^{-3x}(A\cos 5x+B\sin 5x)+e^{-4x}\}^{\frac{1}{4}}}\right)$ **[5]**

**Oct/Nov 2006/1/8**

**44.** Obtain the general solution of the differential equation

$$\frac{d^2y}{dx^2}+6\frac{dy}{dt}+25y=80e^{-3t}.$$ **[5]**

Given that $y=8$ and $\frac{dy}{dt}=-8$ when $t=0$, show that $0\le ye^{3t}\le 10$ for all $t$. $\left(\boldsymbol{y=e^{-3t}(A\cos 4t+B\sin 4t)+5e^{-3t}}\right)$ **[5]**

**May/June 2006/1/8**

**45.** Solve the differential equation

$$\frac{d^2y}{dx^2}+3\frac{dy}{dx}+2y=24e^{2x},$$

given that $y=1$ and $\frac{dy}{dx}=9$ when $x=0$. $\left(\boldsymbol{y=7e^{-x}-6e^{-2x}+2e^{2x}}\right)$ **[7]**

**Oct/Nov 2005/1/4**

**46.** It is given that

$$\frac{d^2y}{dx^2}+(2a-1)\frac{dy}{dx}+a(a-1)y=2a-1+a(a-1)x,$$

where $a$ is a constant. Find $y$ in terms of $a$ and $x$, given that $y$ and $\frac{dy}{dx}$ are both zero when $x=0$. ($\boldsymbol{y=e^{-ax}-e^{(1-a)x}+x}$) **[8]**

Hence show that if $a>1$ then $y\approx x$ as $x\to\infty$. **[2]**

It is given that

$$\frac{d^2z}{dx^2}+(2a-1)\frac{dz}{dx}+a(a-1)z=e^{x},$$

where the constant $a$ is positive. Find $\lim_{x\to\infty} e^{-x}z$. $\left(\frac{\mathbf{1}}{\boldsymbol{a(a+1)}}\right)$ **[3]**

**May/June 2005/1/12O**

**47.** The variable $y$ depends on $x$, and the variables $x$ and $t$ are related by $x = e^t$. Show that

$$x\frac{dy}{dx} = \frac{dy}{dt} \quad and \quad x^2\frac{d^2y}{dx^2} = \frac{d^2y}{dt^2} - \frac{dy}{dt}.$$

**[5]**

(i) Given that $y$ satisfies the differential equation

$$4x^2\frac{d^2y}{dx^2} + 16x\frac{dy}{dx} + 25y = 50(\ln x) - 1,$$

Find a differential equation involving only $t$ and $y$. $\left(\boldsymbol{4\frac{d^2y}{dx^2} + 12\frac{dy}{dt} + 25y = 50t - 1}\right)$ **[2]**

(ii) Show that the complementary function of the differential equation in $t$ and $y$ may be written in the form

$$Re^{-\frac{3}{2}t}\sin(2t + \phi),$$

Where $R$ and $\phi$ are arbitrary constants. $\left(\boldsymbol{R = \sqrt{A^2 + B^2}, \phi = \tan^{-1}\left(\frac{A}{B}\right)}\right)$ **[3]**

(iii) Find a particular integral of the differential equation in $t$ and $y$. $(\boldsymbol{y_p = 2t - 1})$ **[3]**

(iv) Hence find the general solution of the differential equation in $x$ and $y$. $\left(\boldsymbol{y = Rx^{-\frac{3}{2}}\sin(2\ln x + \phi) + 2\ln x - 1}\right)$ **[1]**

**Oct/Nov 2004/1/12E**

**48.** The variable $y$ depends on $x$ and the variables $x$ and $t$ are related by $x = \frac{1}{t}$. Show that

$$\frac{dy}{dx} = -t^2\frac{dy}{dt} \quad and \quad \frac{d^2y}{dx^2} = t^4\frac{d^2y}{dt^2} + 2t^3\frac{dy}{dt}.$$ **[4]**

The variables $x$ and $y$ are related by the differential equation

$$x^5\frac{d^2y}{dx^2} + (2x^4 - 5x^3)\frac{dy}{dx} + 4xy = 14x + 8.$$

Show that

**[2]**

$$\frac{d^2y}{dt^2} + 5\frac{dy}{dt} + 4y = 8t + 14.$$

**[3]**

Hence find the general solution for $y$ in terms of $x$. $\left(\boldsymbol{y = Ae^{-\frac{1}{x}} + Be^{-\frac{4}{x}} + \frac{2}{x} + 1}\right)$

**May/June 2004/1/9**

**49.** Find the general solution of the differential equation

$$\frac{d^2y}{dx^2} + 4\frac{dy}{dt} + 4y = e^{-\alpha t},$$ **[7]**

where $\alpha$ is a constant and $\alpha \neq 2$.

Show that if $\alpha < 2$ then, whatever the initial conditions, $ye^{\alpha t} \to \frac{1}{(2-\alpha)^2}$ as $t \to \infty$. $\left(\boldsymbol{y = (At + B)e^{-2t} + \frac{1}{(\alpha-2)^2}e^{-\alpha t}}\right)$ **[2]**

**Oct/Nov 2003/1/7**

**50.** The variables $x$ and $t$, where $x > 0$ and $0 \leq t \leq \frac{1}{2}\pi$, are related by

$$x\frac{d^2x}{dt^2} + \left(\frac{dx}{dt}\right)^2 + 5x\frac{dx}{dt} + 3x^2 = 3\sin 2t + 15\cos 2t,$$

and the variables $x$ and $y$ are related by $y = x^2$. Show that

$$\frac{d^2y}{dx^2} + 5\frac{dy}{dt} + 6y = 6\sin 2t + 30\cos 2t.$$ **[3]**

Hence find $x$ in terms of $t$, given that $x = 2$ and $\frac{dx}{dt} = -\frac{3}{2}$ when $t = 0$. $\left(\boldsymbol{x^2 = 4e^{-3t} + 3\sin 2t}\right)$ **[10]**

**May/June 2003/1/9**

**51.** The value of the assets of a large commercial organization at time $t$, measured in years, is \$$(10^8 Y + 10^9)$. The variables $y$ and $t$ are related by the differential equation

$$\frac{d^2y}{dt^2} + 5\frac{dy}{dt} + 6y = 15\cos 3t - 3\sin 3t.$$

Find $y$ in terms of $t$ given that $y = 3$ and $\frac{dy}{dt} = -2$ when $t = 0$. $(\boldsymbol{y = 4e^{-2t} - e^{-3t} + \sin 3t})$ **[9]**

Show that, for large values of $t$, the value of the assets is \$$9.5 \times 10^8$ for about a third of the time. **[3]**

**Oct/Nov 2002/1/8**

**52.** Find the general solution of the differential equation

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 5y = 15x + 16.$$ **[6]**

Show that, whatever the initial conditions, $y \approx 3x + 2$ when $x$ is large and positive. $(\boldsymbol{y = e^{-x}(A\cos 2x + B\sin 2x) + 3x + 2})$ **[1]**

**May/June 2002/1/4**

**53.** Find, in the form $y = f(x)$, the solution of the differential equation $\frac{dy}{dx} + y\tanh x = 2\cosh x$, given that $y = \frac{3}{4}$ when $x = \ln 2$. $\left(\boldsymbol{y = \frac{x - \ln 2}{\cosh x} + \sinh x}\right)$ **[8]**

**CAIE 9795/May/June 2018/1/5**

**54.** (a) (i) Use the result

$$\tanh^{-1} x = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right)$$

to find $\frac{d}{dx}(\tanh^{-1} x)$. $\left(\boldsymbol{\frac{1}{1-x^2}}\right)$ **[2]**

(ii) Find $\frac{d}{dx}[\ln(\tanh^{-1} x)]$. $\left(\boldsymbol{\frac{1}{(1-x^2)\tanh^{-1} x}}\right)$ **[1]**

(iii) Hence, or otherwise, find the general solution of

$$\frac{dy}{dx} + \left[\frac{2}{(1-x^2)\tanh^{-1} x}\right]y = \frac{3x^2}{(\tanh^{-1} x)^2}.$$ **[4]**

$\left(\boldsymbol{(\tanh^{-1} x)^{-2}(x^3 + c)}\right)$

**Nov 1997/1/5 (part)**

**55.** (a) Obtain the solution, for $y$ in terms of $x$, of the differential equation

$$\frac{dy}{dx} - (4\tanh x)y = \sinh 2x,$$

given that $y = 0$ when $x = 0$. $\left(\boldsymbol{y = \frac{1}{4}\sinh^2 2x}\right)$ **[6]**

**Nov 1995/1/5 (part)**

**56.** (a) Find the general solution of the differential equation

$$(\coth x + 2\operatorname{cosech} x)\frac{dy}{dx} + y = x$$

where $x \neq 0$. $(\boldsymbol{y(2 + \cosh x) = x\cosh x - \sinh x + c})$ **[5]**

**June 1994/1/5 (part)**

**57.** (i) Obtain the solution of the differential equation

$$\frac{dy}{dx} + (\tanh x)y = 1$$

such that $y = 1$ when $x = 0$. $(\boldsymbol{y = \tanh x + \operatorname{sech} x})$ **[6]**

**June 1988/1/5 (part)**

**58.** Prove that

(a) $\displaystyle\int \operatorname{sech} x\, dx = \tan^{-1}(e^x) + C,$

(b) $\int \operatorname{sech}^3 x\,dx = \frac{1}{2}\tanh x \operatorname{sech} x + \tan^{-1}(e^x) + C.$

where $C$ is, in each case, an arbitrary constant.

Find the particular solution of the differential equation $\frac{dy}{dx} - 3y\tanh x = 1$ for which $y = 0$ when $x = 0$. $\boldsymbol{\left(y = \frac{1}{2}\sinh x\cosh x + \cosh^3 x\left(\tan^{-1} e^x - \frac{\pi}{4}\right)\right)}$

**MB November 1979/1/22**